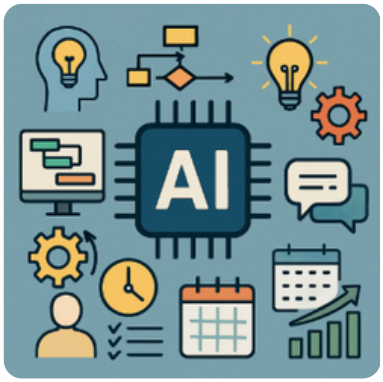
Project Report

Smart Attendance System Using Python (Image Capture Based)



Program Name : BCA(Data Science)

Subject Name : Artificial Intelligence Lab
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Teacher's Signature

Acknowledgement

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Smart Attendance System Using Python
(Image Capture Based)

1.Abstract

The Smart Attendance System is an innovative approach to automate the process of attendance marking using modern technology. Traditional attendance systems are time-consuming and prone to human errors. This project uses Python programming and computer vision techniques to capture an image of the user through a webcam and store attendance records in a digital format.
The system captures a single image, enhances its quality using image preprocessing techniques, and records attendance in a CSV file along with time. The captured image is also saved for verification purposes. This project demonstrates the application of Artificial Intelligence concepts in real-life scenarios.

2. Introduction

Attendance management is an essential activity in educational institutions and workplaces. Traditionally, attendance is recorded manually using registers or sheets, which is time-consuming, error-prone, and inefficient. With the advancement in technology, automated systems are being developed to simplify such processes.
This project presents a Smart Attendance System using Python and OpenCV, which captures a user's image through a webcam and records attendance digitally in a CSV file. The system enhances image quality using preprocessing techniques and stores both attendance data and captured images in a structured manner. This project demonstrates the practical implementation of Artificial Intelligence concepts like computer vision and image processing in real life.

3. Objectives

- To develop an automated attendance system
- To capture real-time images using a webcam
- To improve image quality using preprocessing techniques
- To store attendance records digitally
- To reduce manual errors and paperwork
- To create a simple and user-friendly system

4. Problem Statement

- Manual attendance systems have several drawbacks:
- Time-consuming process
- Chances of proxy attendance
- Human errors in recording
- Difficult to maintain records
- This project aims to solve these problems by providing a simple automated attendance solution.

5. Literature Review

Various attendance systems have been developed over time, such as:

- Manual register-based systems
- RFID-based systems
- Biometric systems (fingerprint, iris)
- Face recognition systems

However, these systems may be costly or complex. This project provides a simple and cost-effective alternative using Python.

6. System Requirements

Hardware:

- Laptop/Computer
- Webcam

Software:

- Python 3.x
- Code Editor (IDLE / VS Code)

7. Tools and Technologies

1. **Python**

- A high-level programming language used to implement the system.

2. **OpenCV (Computer Vision)**

Used for:

- Capturing image from webcam
- Image processing
- Displaying output

3. **Pandas**

Used for:

- Reading and writing CSV files
- Managing attendance data

4. **Datetime Module**

Used to:

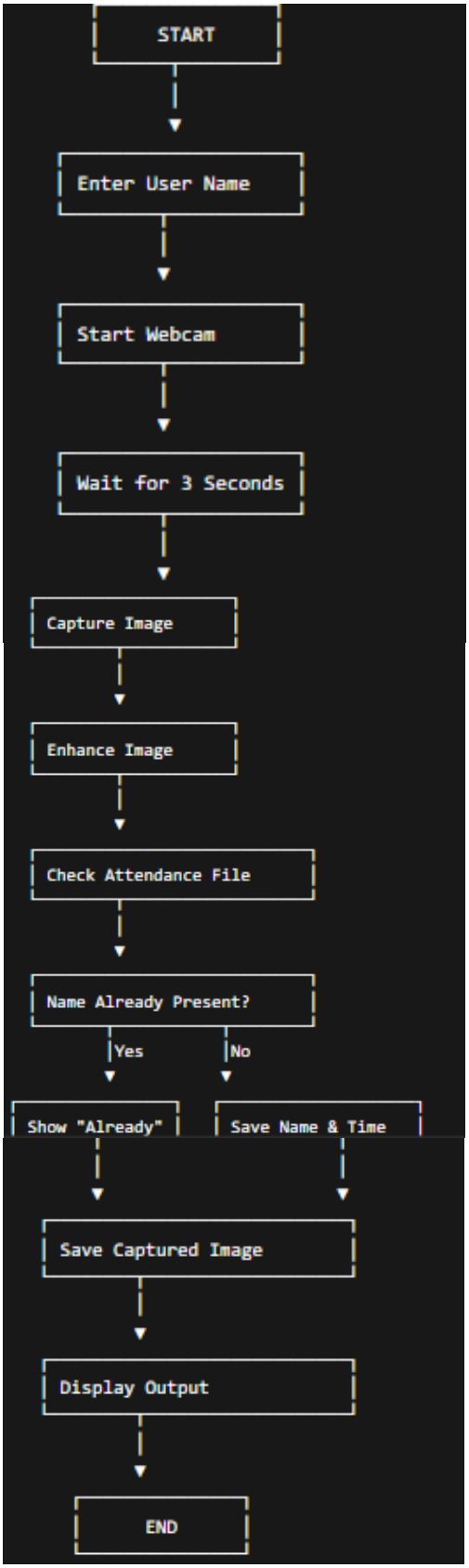
- Record current time of attendance

5. **OS Module**

Used to:

- Create folders
- Manage file paths

8.Flowchart


START
Enter User Name
Start Webcam
Wait for 3 Seconds
Capture Image
Enhance Image
Check Attendance File
Name Already Present?
Yes
No
Show "Already"
Save Name & Time
Save Captured Image
Display Output
END

9. Working

Step 1: User Input
The user enters their name through the terminal.

Step 2: Camera Activation
The system activates the webcam and waits for 3 seconds to allow the user to adjust.

Step 3: Image Capture
A single frame is captured from the webcam.

Step 4: Image Enhancement
- The captured image is enhanced using:
- Brightness adjustment
- Contrast improvement
- Smoothing

Step 5: Attendance Recording
The system checks if the name already exists
If not, it stores:
Name
Time

Step 6: File Storage
- Attendance saved in CSV file
- Image saved in folder

Step 7: Output Display
The captured image is shown for 3 seconds and then the system closes.

10.System Architecture

User → Webcam → Image Capture → Image Processing → Attendance File → Output

11. Algorithm

1. Start
2. Input user name
3. Open webcam
4. Wait for 3 seconds
5. Capture image
6. Enhance image brightness
7. Check attendance file
8. If name not present:
   Save name and time
9. Save image
10. Display output
11. Stop

12.Implementation

The system is implemented using Python programming. OpenCV is used to capture images from the webcam. The image is enhanced using brightness and contrast adjustment. Pandas is used to store attendance in CSV format.
The system also creates a folder to store attendance records and images.

13.Code

```
import cv2
import pandas as pd
from datetime import datetime
import time
import os

# Folder path
folder_path = r"C:/Users/vikit/Desktop/attendence_marked"

# Create folder if not exists
if not os.path.exists(folder_path):
    os.makedirs(folder_path)

# File path
file_name = os.path.join(folder_path, "attendance.csv")

# Start camera
cap = cv2.VideoCapture(0)

name = input("Enter your name: ")

print("Camera starting... Adjust yourself in light")
time.sleep(3)

ret, frame = cap.read()

if not ret:
    print("Camera error")
    cap.release()
    exit()

# Improve brightness & quality
frame = cv2.convertScaleAbs(frame, alpha=1.5, beta=40)  # brightness
frame = cv2.GaussianBlur(frame, (5, 5), 0)  # smooth image

# Show captured image
cv2.imshow("Captured Photo", frame)

# Create CSV if not exists
if not os.path.exists(file_name):
    df = pd.DataFrame(columns=["Name", "Time"])
    df.to_csv(file_name, index=False)
```

```
# Load file
df = pd.read_csv(file_name)

# Mark attendance
if name not in df["Name"].values:
    now = datetime.now()
    time_now = now.strftime("%H:%M:%S")

    new_entry = pd.DataFrame([[name, time_now]], columns=["Name", "Time"])
    df = pd.concat([df, new_entry], ignore_index=True)
    df.to_csv(file_name, index=False)

    print("Attendance Marked ")
else:
    print("Already Marked ")

# Save photo in folder
image_path = os.path.join(folder_path, f"{name}.jpg")
cv2.imwrite(image_path, frame)

print("Photo saved in folder ")

cv2.waitKey(3000)

cap.release()
cv2.destroyAllWindows()
```

14. Output

Terminal Output:

User enters name and receives confirmation message

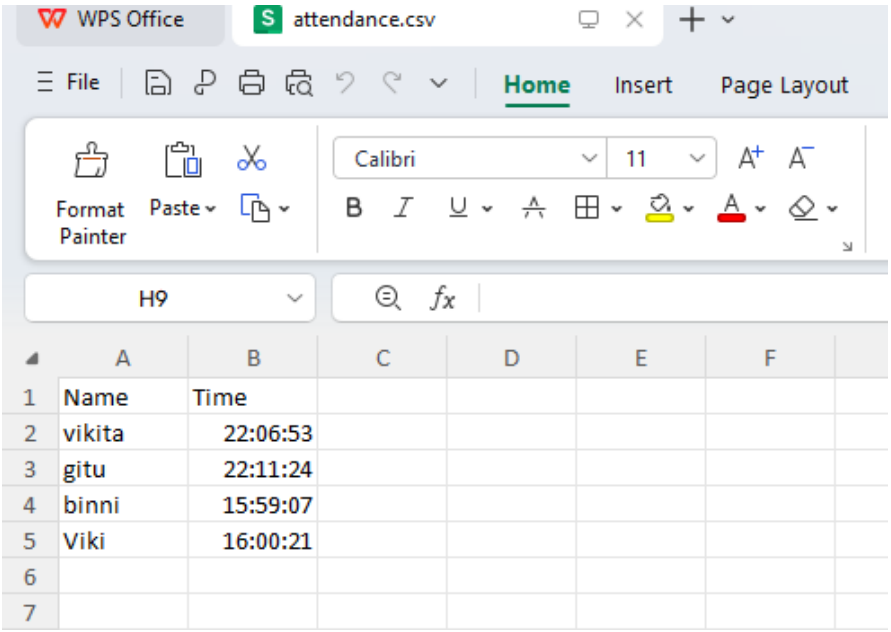
```
==================================================
Enter your name: Viki
Camera starting... Adjust yourself in light
Attendance Marked
Photo saved in folder
```

Image Output:

Captured image is displayed and saved.

File Output:

Attendance is stored in CSV file.



	A	B	C	D	E	F
1	Name	Time				
2	vikita	22:06:53				
3	gitu	22:11:24				
4	binni	15:59:07				
5	Viki	16:00:21				
6						
7						

15. Result

- The Smart Attendance System was successfully developed and tested.
- The system captures images using a webcam and records attendance in a CSV file.
- The output is generated in three forms: terminal messages, captured images, and attendance records.
- The system is simple and user-friendly, making it easy to operate.
- Attendance is recorded accurately without duplication, as the system checks existing entries.
- The processing time is very fast, and the entire operation completes within a few seconds.
- Image enhancement techniques improve brightness and clarity of the captured image.
- The system works best under proper lighting conditions.
- In low-light conditions, the image quality may decrease, affecting output clarity.
- The system depends on the quality of the webcam for better results.
- Overall, the system is efficient, reliable, and reduces manual effort in attendance management.

16. Advantages

- Reduces manual effort as attendance is recorded automatically
- Saves time compared to traditional register-based systems
- Easy to use and understand (user-friendly system)
- Provides digital storage of attendance data
- Reduces chances of human error and duplication
- Cost-effective (requires only a computer and webcam)
- Data can be easily accessed and managed
- Improves efficiency and accuracy in attendance management
- No need for physical paperwork

17. Limitations

- Depends on proper lighting conditions for good image quality
- Performance depends on webcam quality
- No automatic face recognition (manual name entry required)
- Possibility of proxy attendance still exists
- Works for only one user at a time
- Limited security features
- Not suitable for large-scale systems without upgrades

18. Future Scope

- Integration of face recognition for automatic identification
- Development of a graphical user interface (GUI)
- Use of database (MySQL, SQLite) instead of CSV file
- Cloud-based attendance system for remote access
- Mobile application integration
- Multi-user and real-time attendance system
- Addition of security features like authentication
- Integration with biometric systems (fingerprint, iris)

19. Applications

- Schools and colleges for student attendance
- Offices and companies for employee attendance
- Coaching institutes and training centers
- Workshops, seminars, and events
- Small organizations and startups
- Any place where attendance tracking is required

20.Conclusion

- The Smart Attendance System successfully provides an automated solution for recording attendance.
- The system captures images using a webcam and stores attendance data in a digital format, reducing the need for manual registers.
- It helps in saving time and effort for both users and administrators.
- The system ensures better accuracy by avoiding duplicate entries and minimizing human errors.
- It is simple, user-friendly, and easy to implement, making it suitable for beginners and small-scale applications.
- The use of Python and image processing techniques demonstrates the practical application of Artificial Intelligence concepts in real-life scenarios.
- The system maintains organized records in a CSV file, which can be easily accessed and managed.
- Although the system has some limitations such as dependency on lighting and lack of face recognition, it still performs efficiently for basic attendance management.
- The project can be further improved by adding advanced features like face recognition, database integration, and GUI.
- Overall, the Smart Attendance System is an effective, reliable, and cost-efficient solution for modern attendance management.

21. References

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    - Attendance system development